EPIC Meeting

Translations Exclusive Victim Cache Summary

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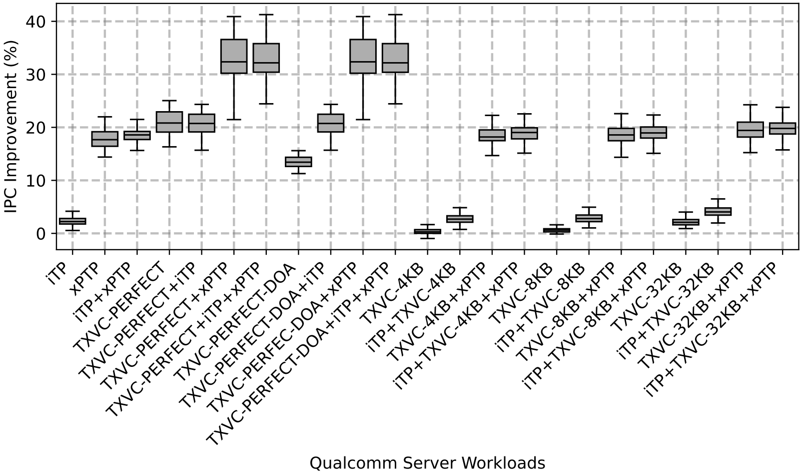
Barcelona Supercomputing Center (BSC)

Content

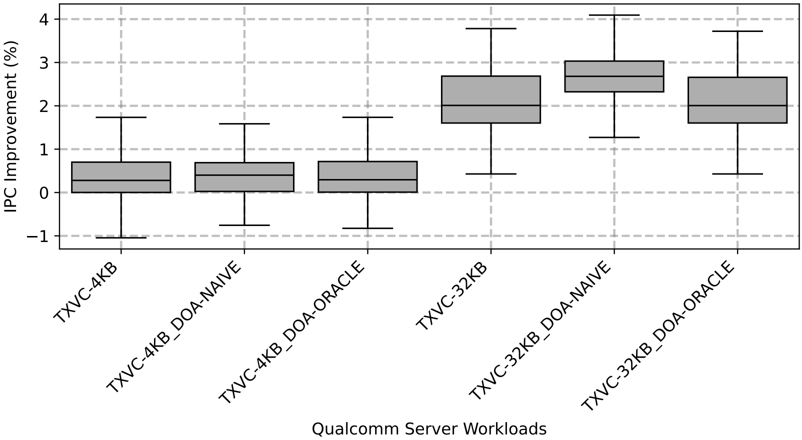
Performance of iTP, xPTP and L1D-TXVC



Summary

- Unlimited storage for TXVC improves performance
- Combined with xPTP provided additional performance
- Budgeted version offer negligible performance
- Need better utilization of TXVC to achieve any significant performance benefits

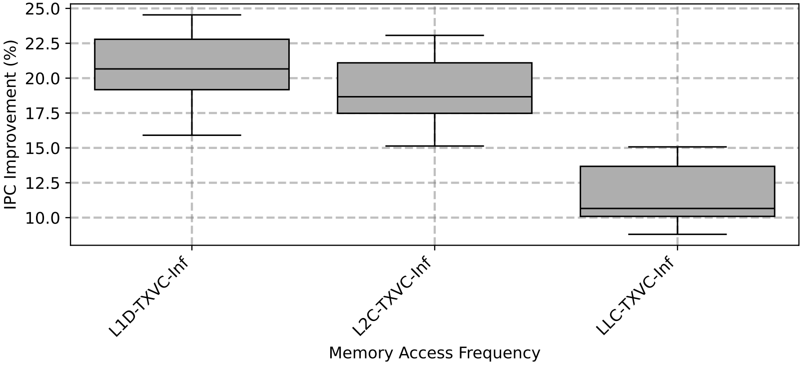
Performance of TXVC with DOA predictor



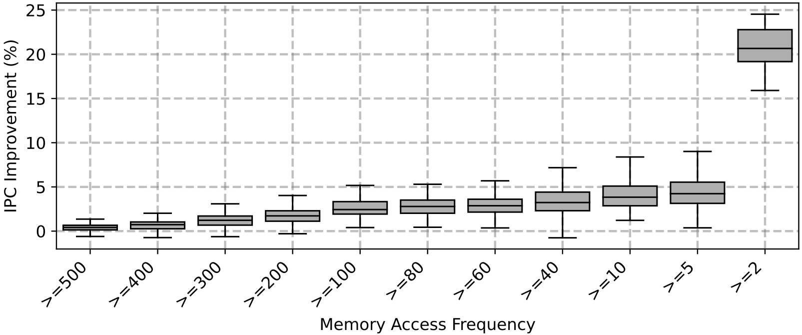
Summary

- Naive performs very well
- Cannot achieve significant performance gains

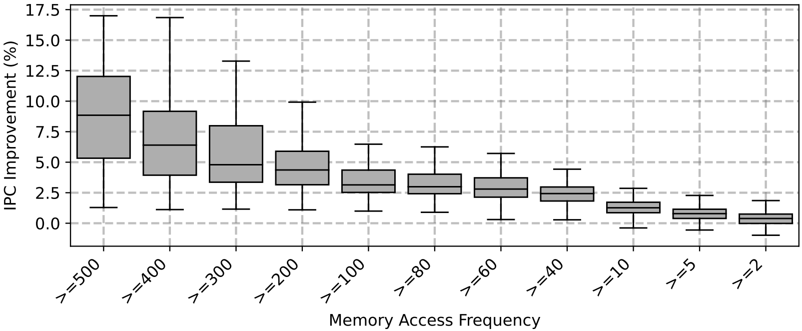
Which cache level benefits most from TXVC?



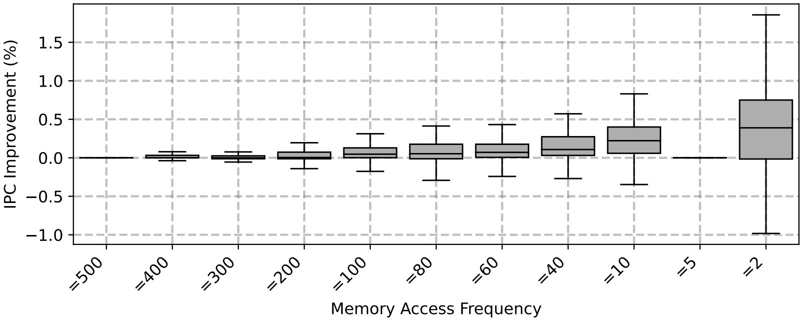
L1D-TXVC memory access frequency analysis



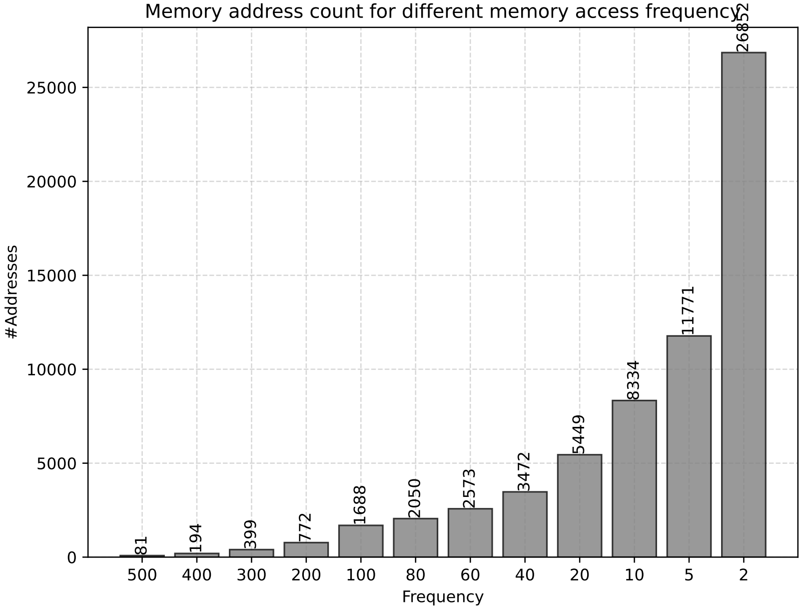
L1D-TXVC memory access frequency analysis



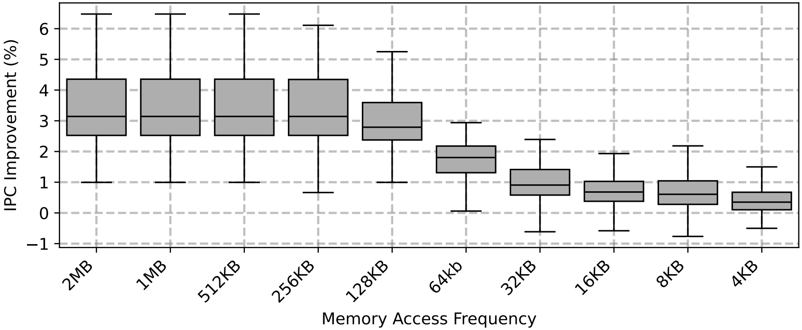
L1D-TXVC memory access frequency analysis



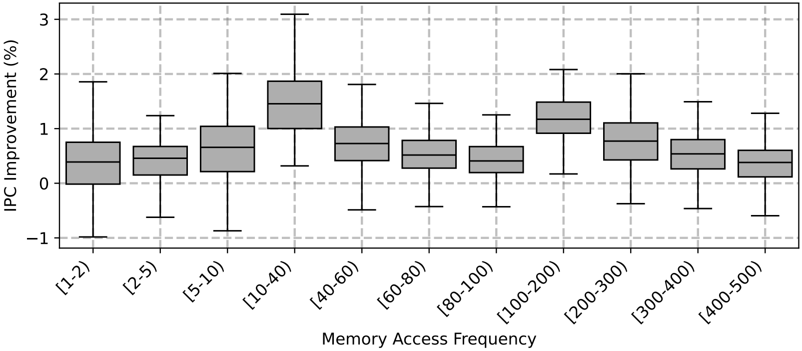
L1D-TXVC memory access frequency analysis



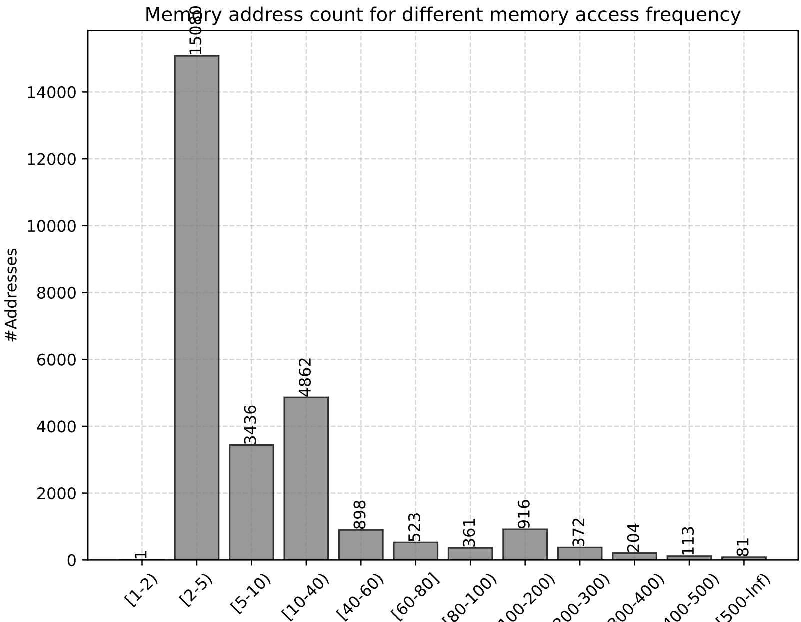
L1D-TXVC size analysis for freq <= 100



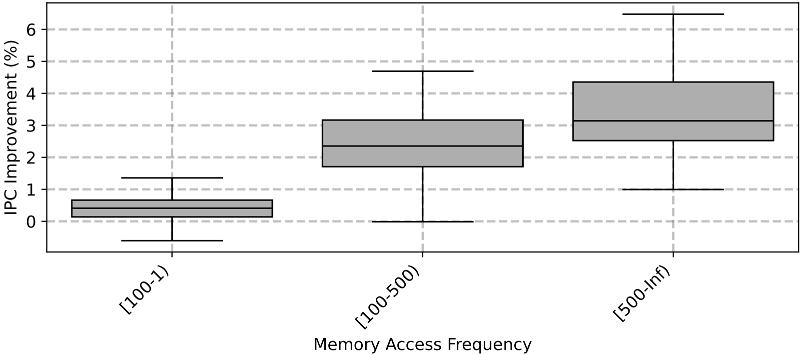
L1D-TXVC size analysis for freq ranges (1/2)



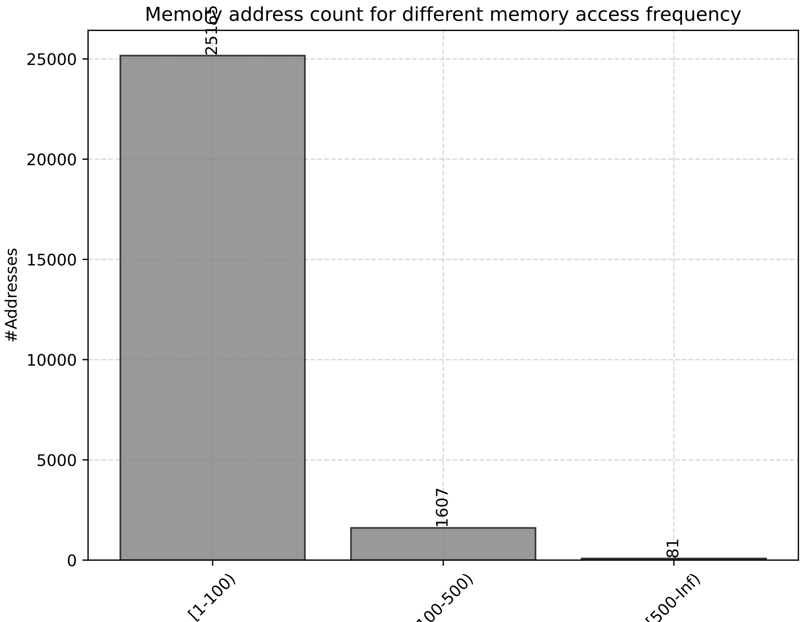
L1D-TXVC size analysis for freq ranges (1/2)



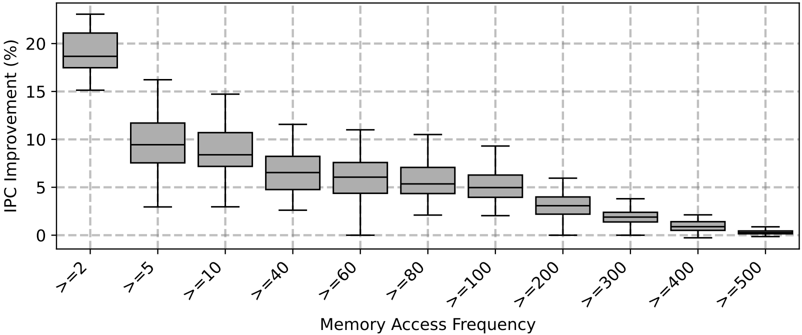
L1D-TXVC size analysis for freq ranges (2/2)



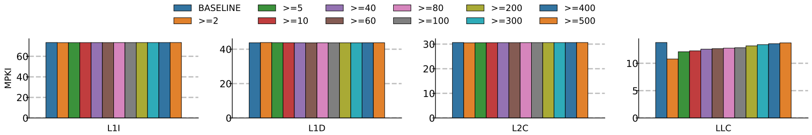
L1D-TXVC size analysis for freq ranges (2/2)



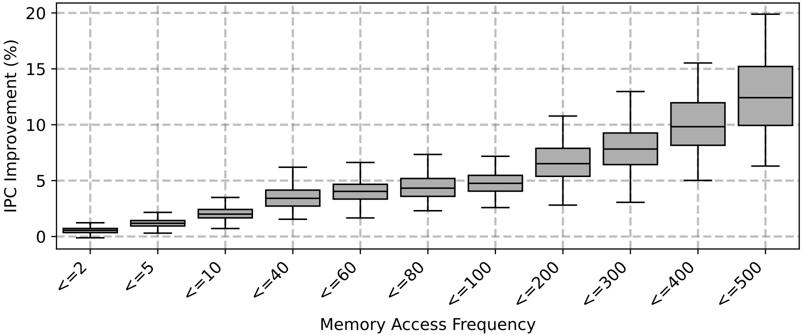
L2C-TXVC memory access frequency analysis



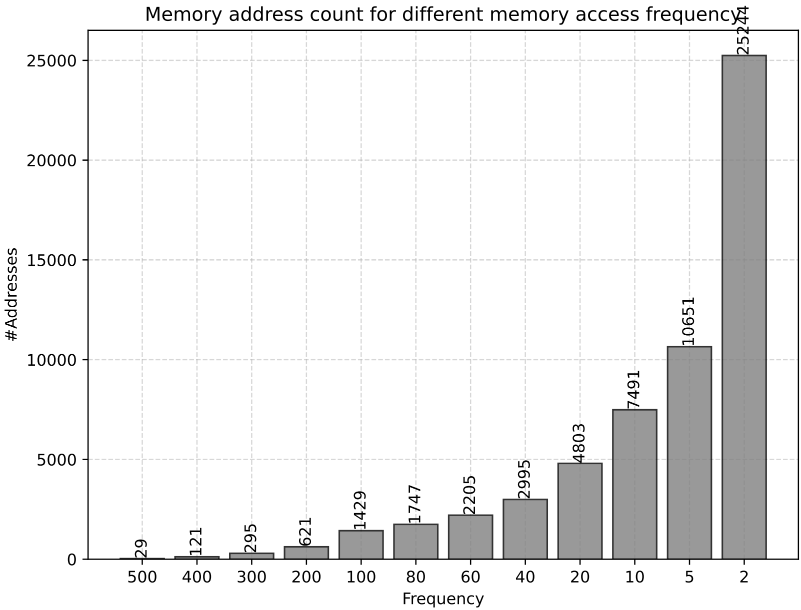
L2C-TXVC memory access frequency analysis



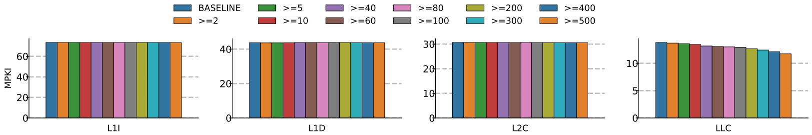
L2C-TXVC memory access frequency analysis



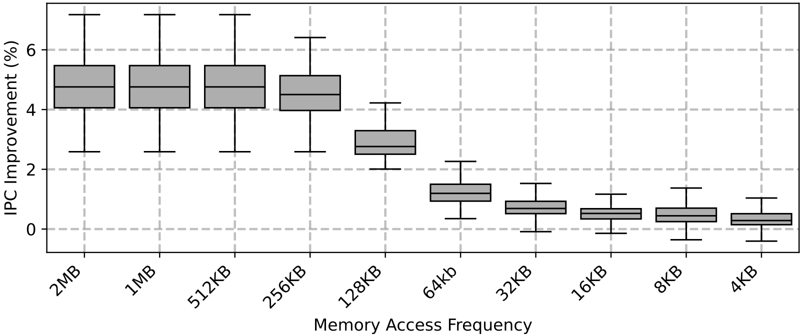
L2C-TXVC memory access frequency analysis



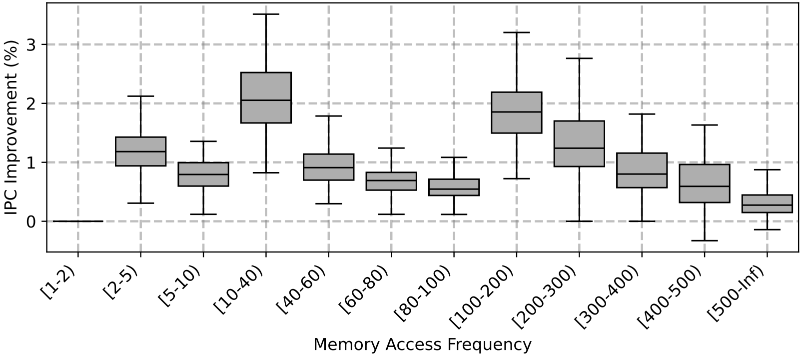
L2C-TXVC memory access frequency analysis



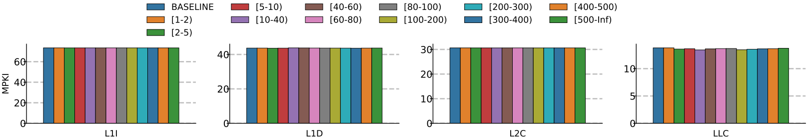
L2C-TXVC size analysis for freq $<=$ 100



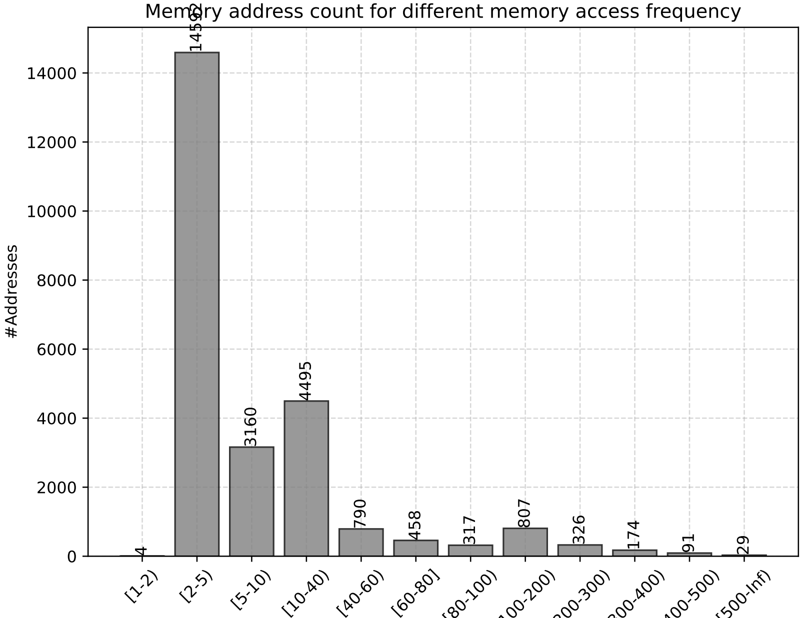
L2C-TXVC size analysis for freq ranges (1/3)



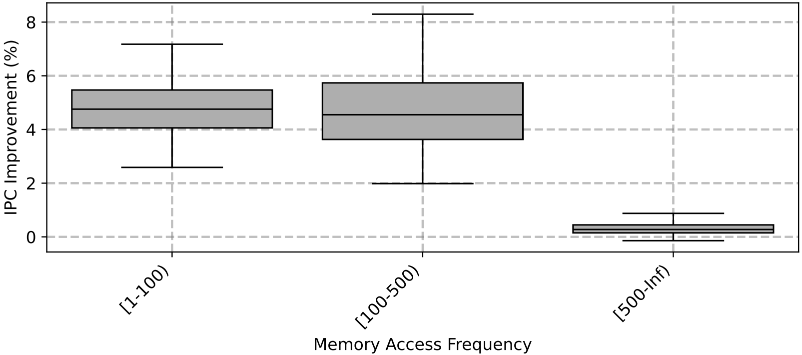
L2C-TXVC size analysis for freq ranges (1/3)



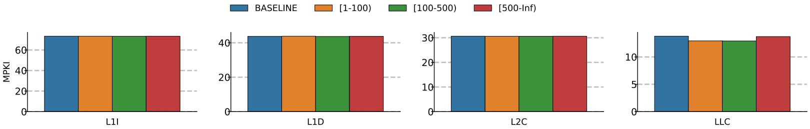
L2C-TXVC size analysis for freq ranges (1/3)



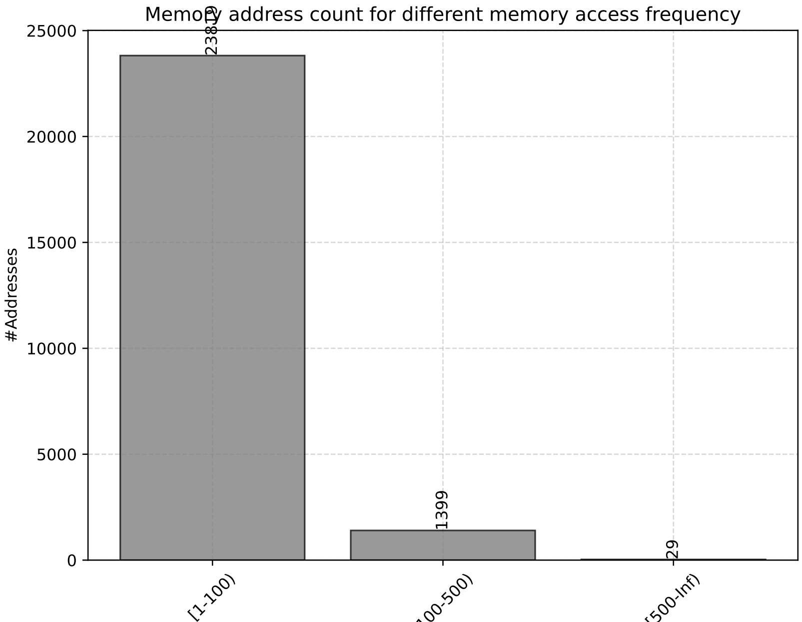
L2C-TXVC size analysis for freq ranges (2/3)



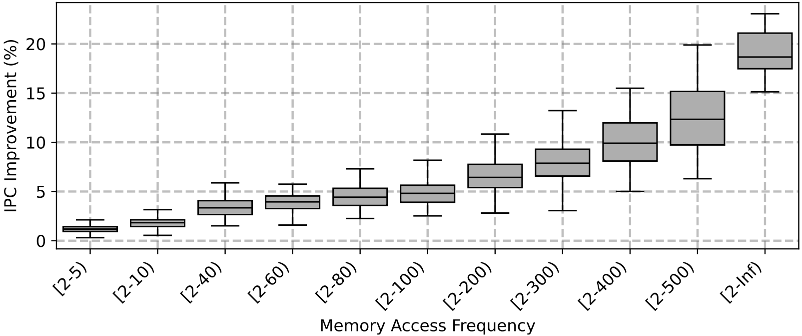
L2C-TXVC size analysis for freq ranges (2/3)



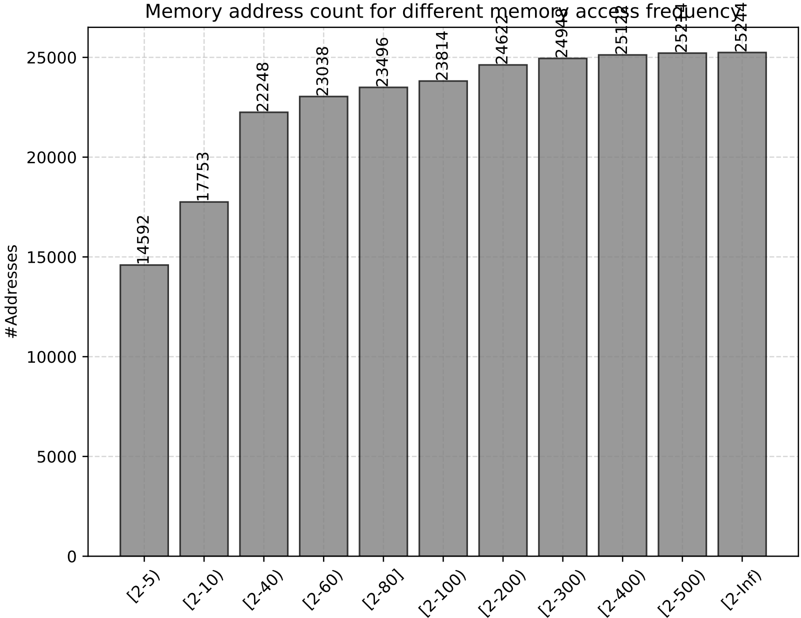
L2C-TXVC size analysis for freq ranges (2/3)



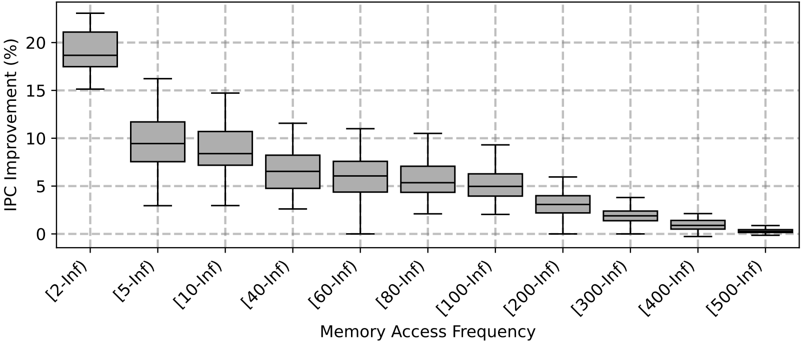
L2C-TXVC size analysis for freq ranges (3/4)



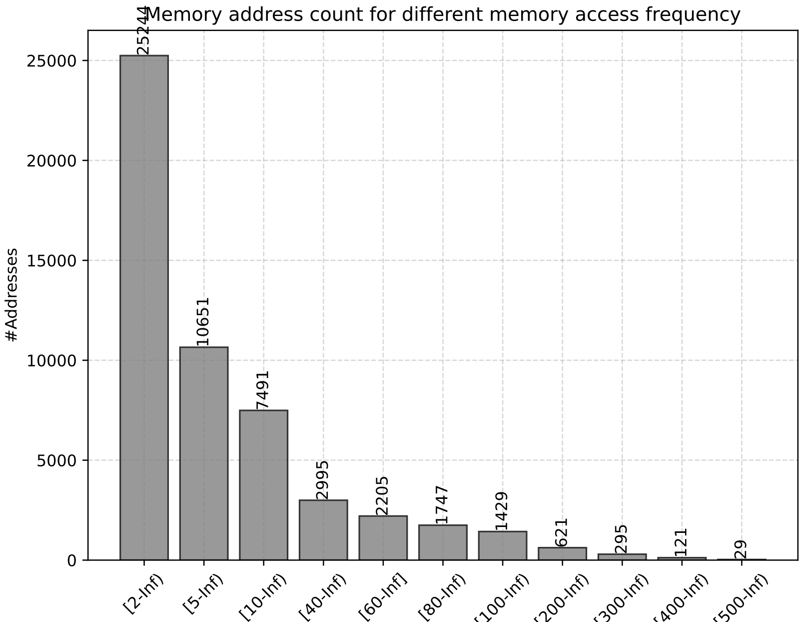
L2C-TXVC size analysis for freq ranges (3/4)



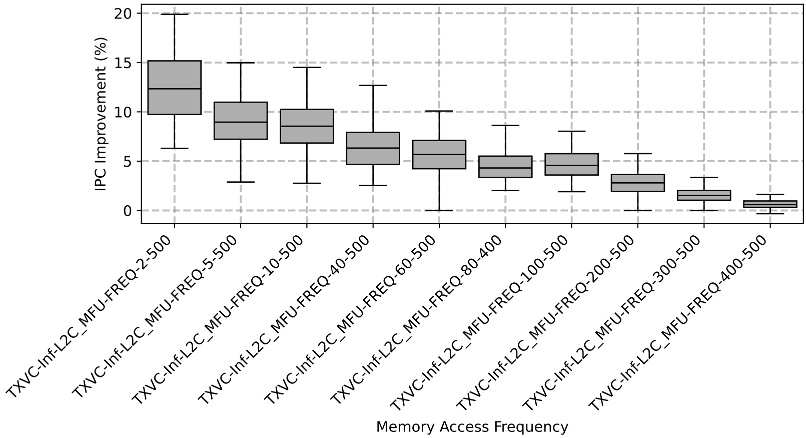
L2C-TXVC size analysis for freq ranges (4/4)



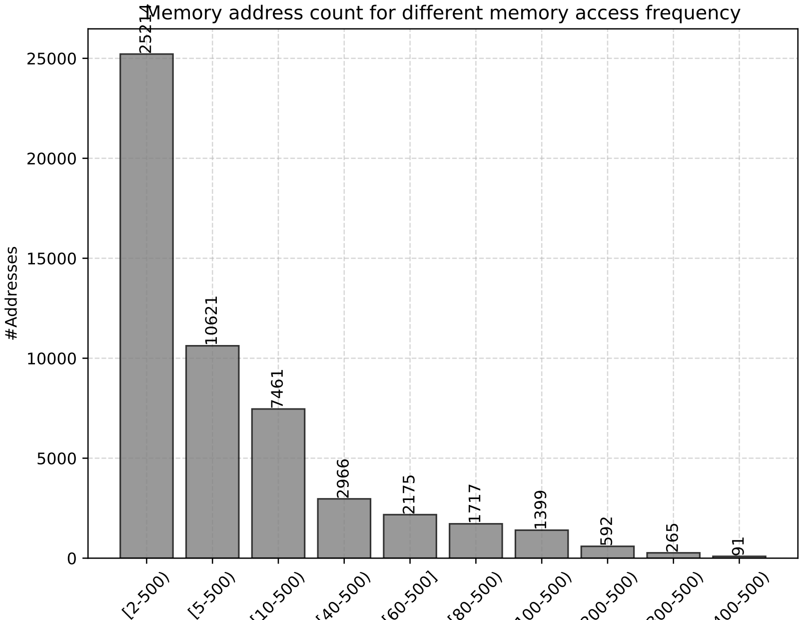
L2C-TXVC size analysis for freq ranges (4/4)



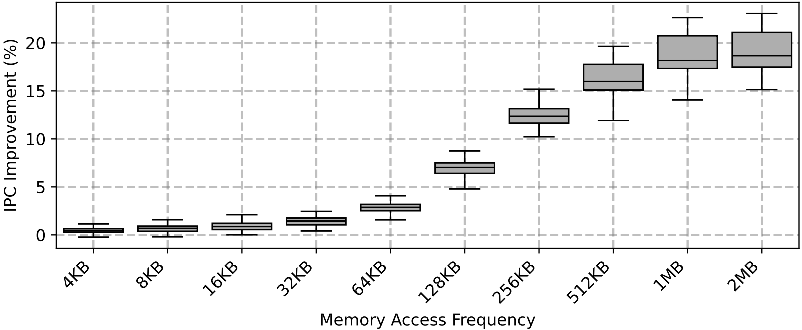
L2C-TXVC size analysis for freq ranges (*)



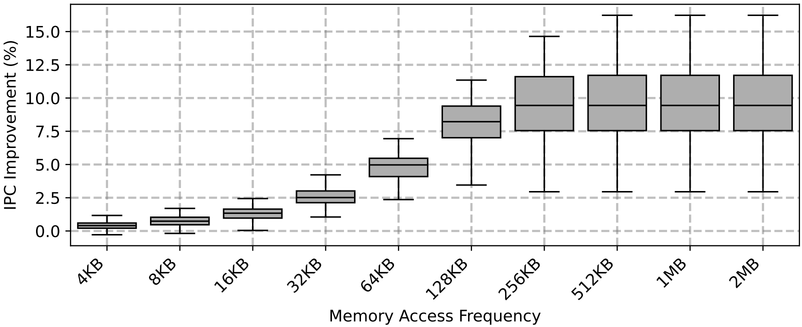
L2C-TXVC size analysis for freq ranges (*)



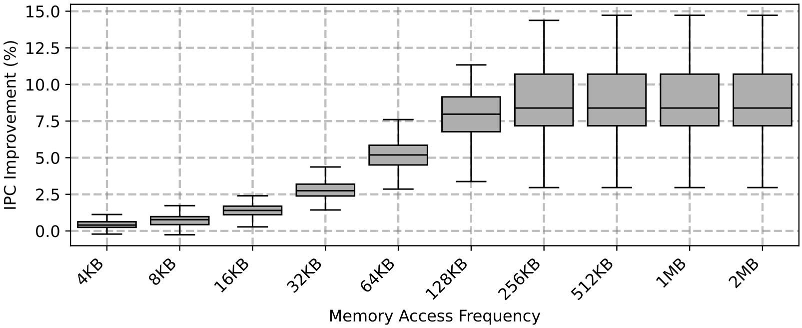
L2C-TXVC size impact analysis for range 2-Inf



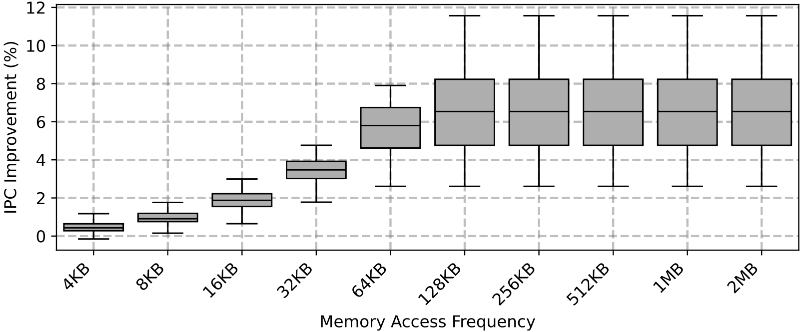
L2C-TXVC size impact analysis for range 5-Inf



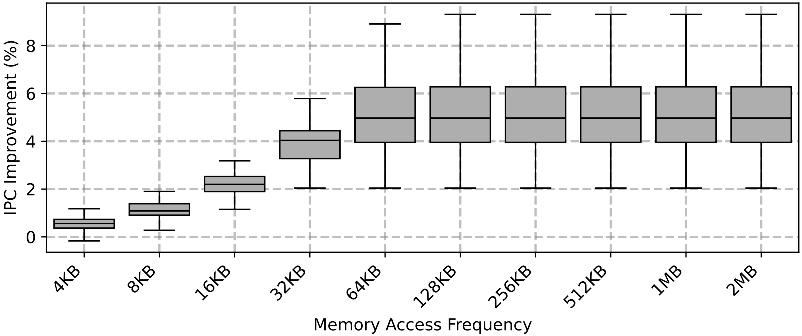
L2C-TXVC size impact analysis for range 10-Inf



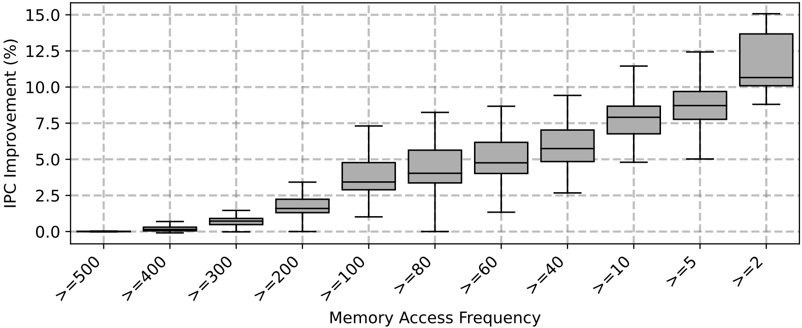
L2C-TXVC size impact analysis for range 40-Inf



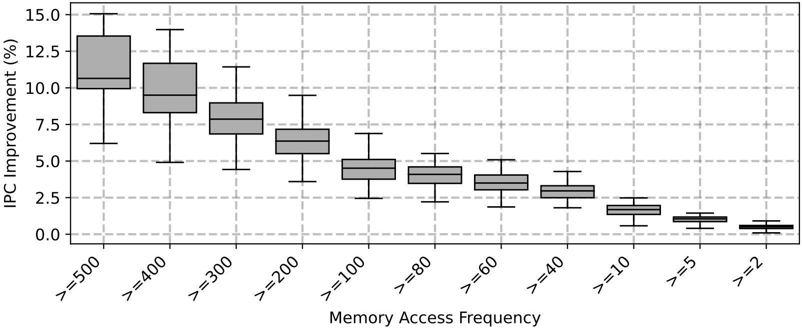
L2C-TXVC size impact analysis for range 100-Inf



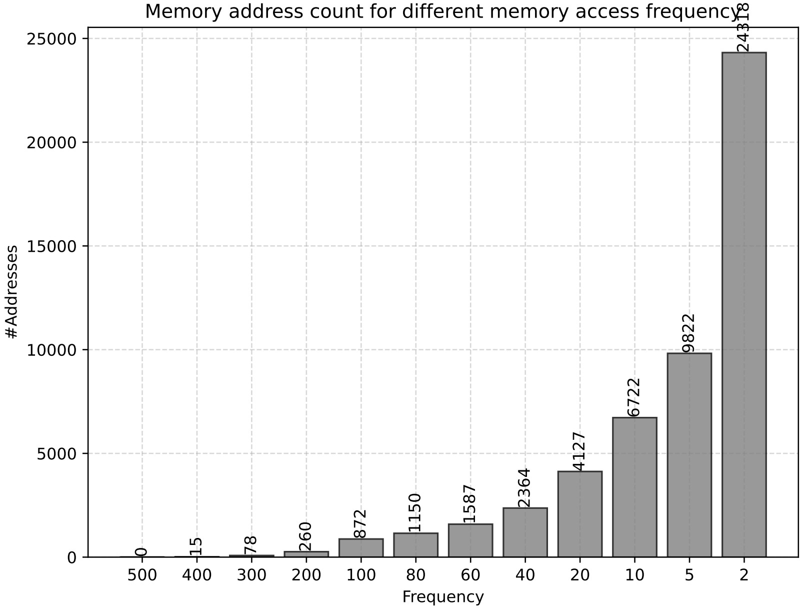
LLC-TXVC memory access frequency analysis



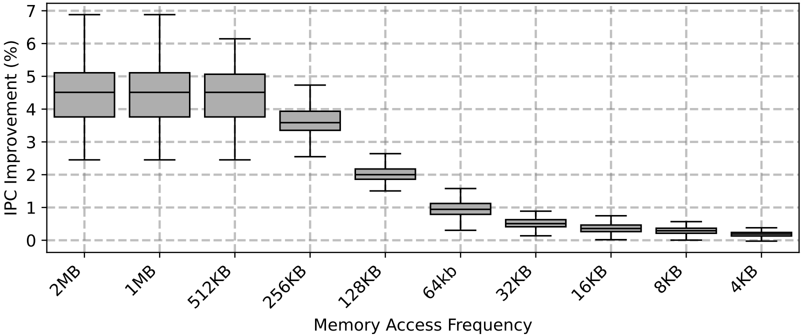
LLC-TXVC memory access frequency analysis



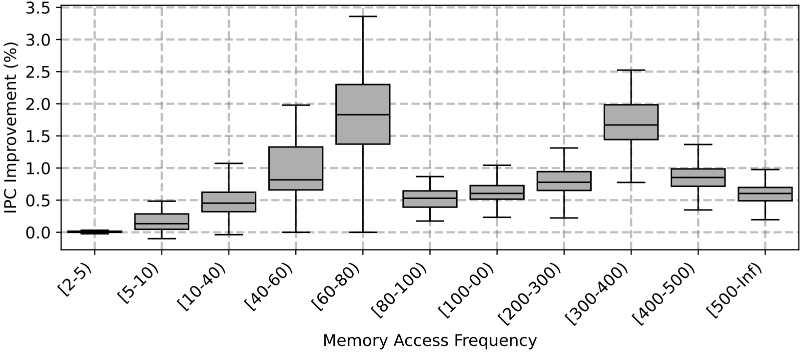
LLC-TXVC memory access frequency analysis



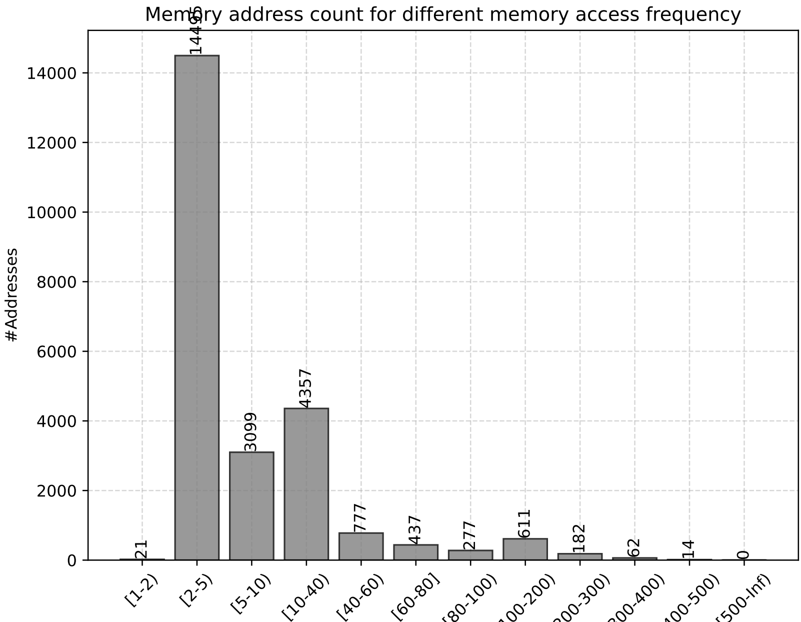
LLC-TXVC size analysis for freq <= 100



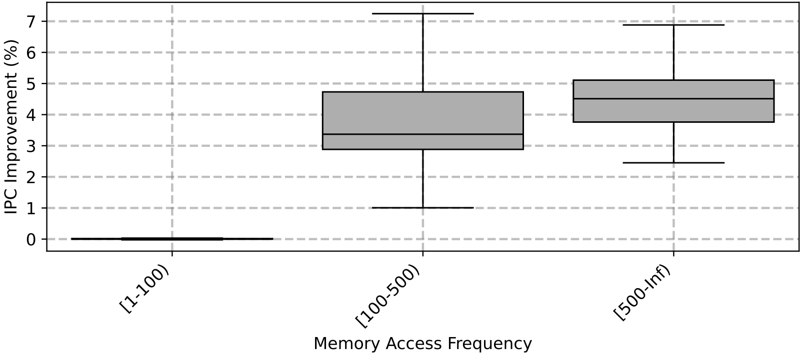
LLC-TXVC size analysis for freq ranges (1/2)



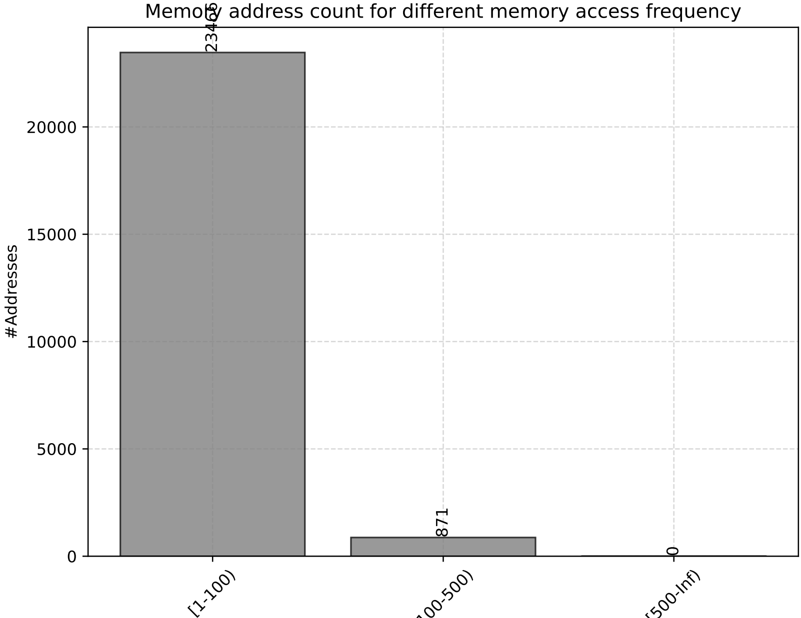
LLC-TXVC size analysis for freq ranges (1/2)



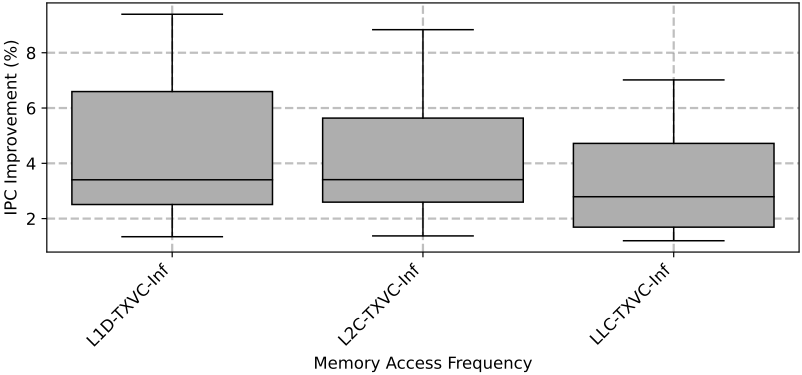
LLC-TXVC size analysis for freq ranges (2/2)



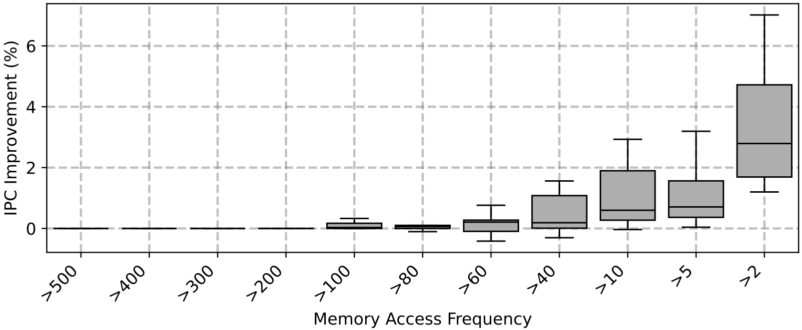
LLC-TXVC size analysis for freq ranges (2/2)



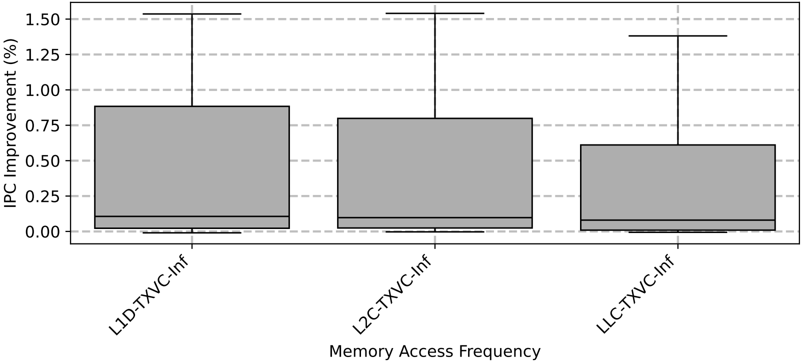
Other Server Workloads Performance



Other Server Workloads Performance

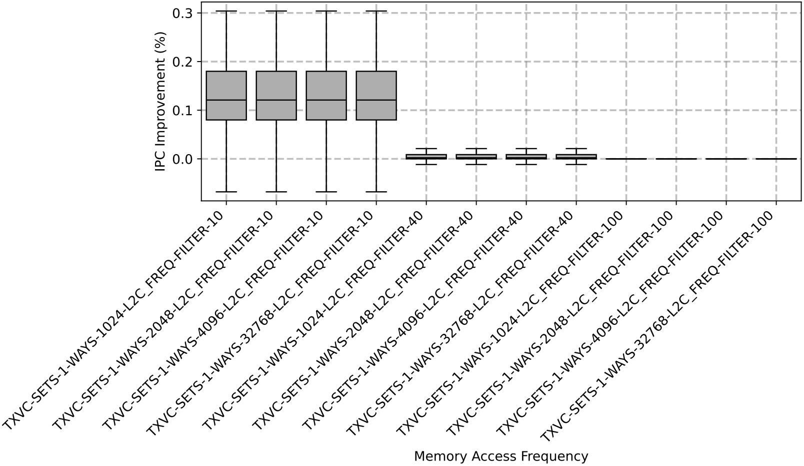


Other SPEC CPU 2017 Workloads Performance

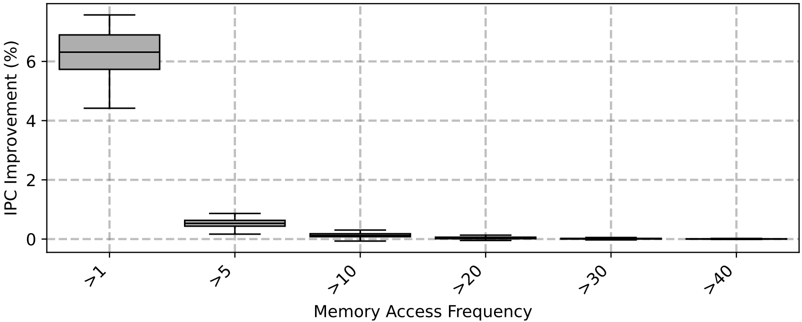


Filter Analysis

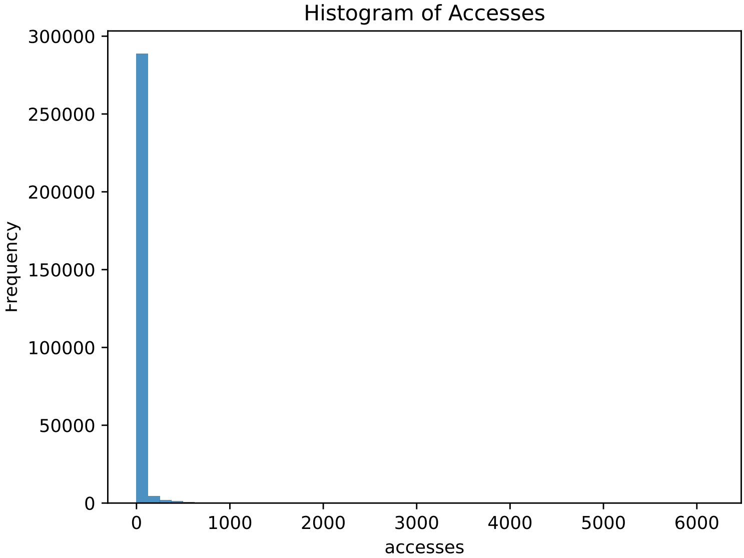
Filter effectiveness over Frequencies - recap



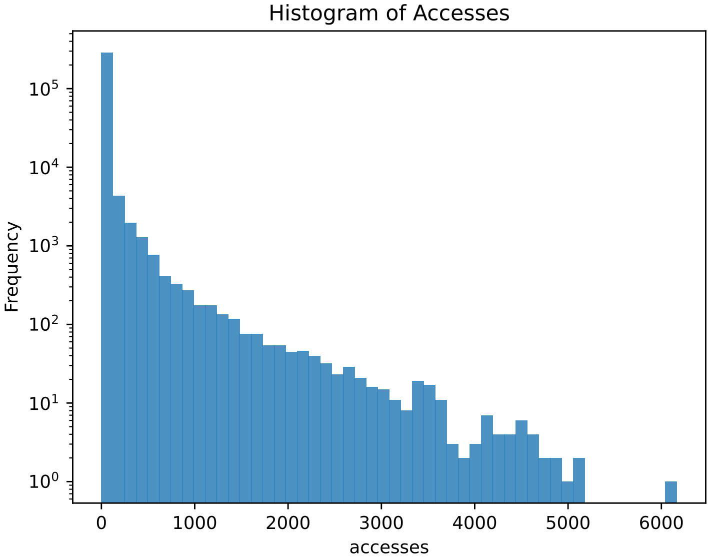
Exploring lower access frequencies



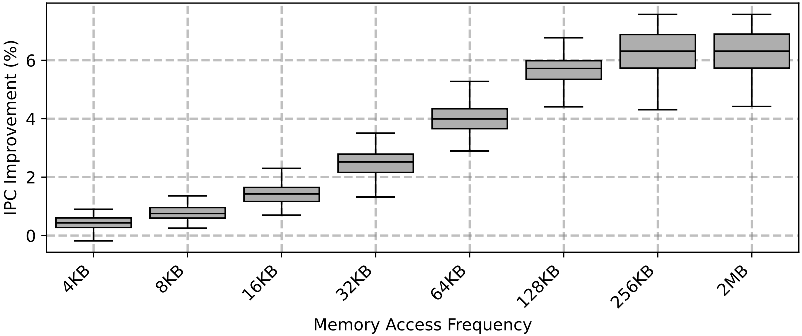
Real Access Frequency of L2C (linear)



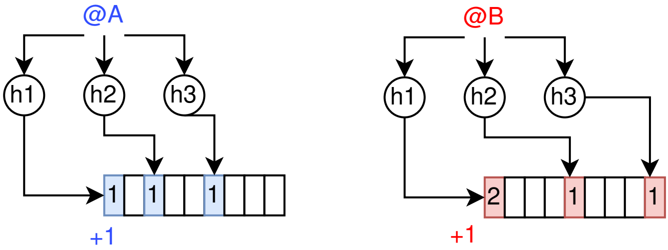
Real Access Frequency of L2C (log)



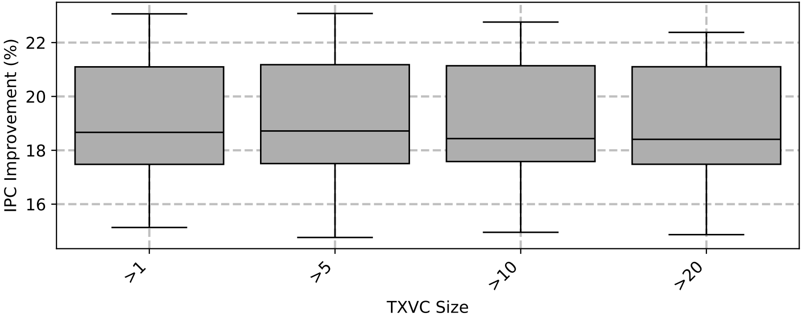
TXVC Size impact



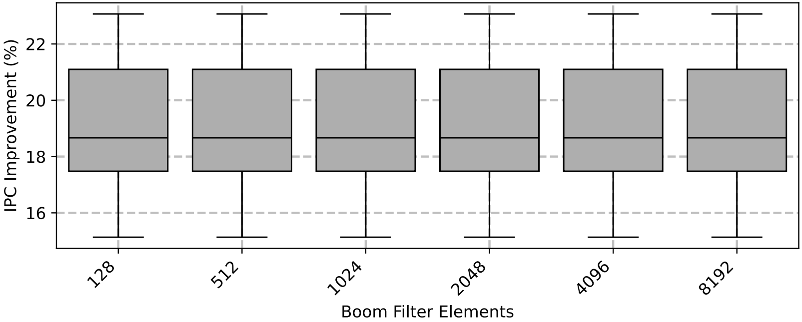
Bloom Frequency Filter (BFF)



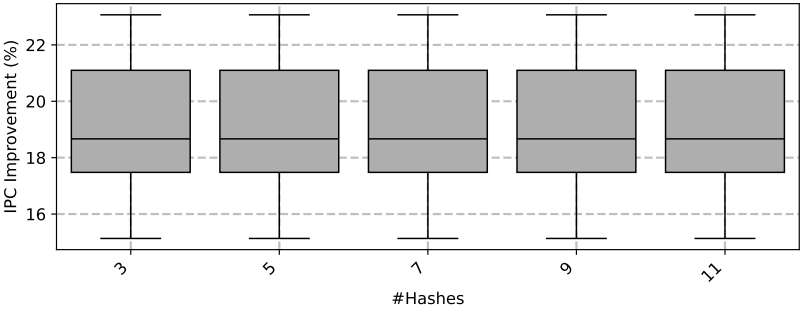
BFF Threshold Impact



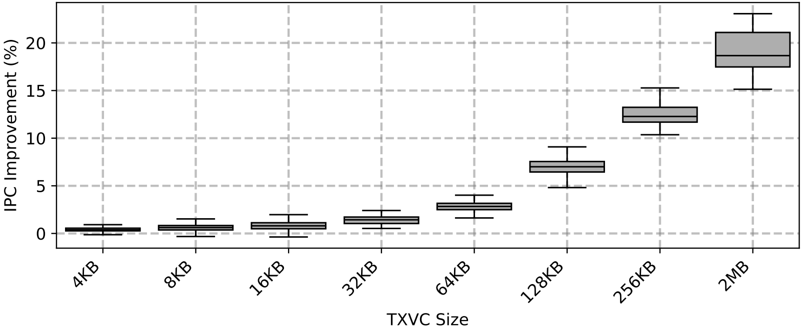
BFF Size Impact



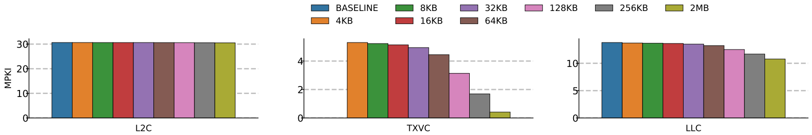
BFF Number of Hashes Impact



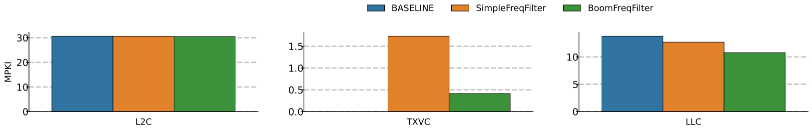
BFF TXVC Size Impact



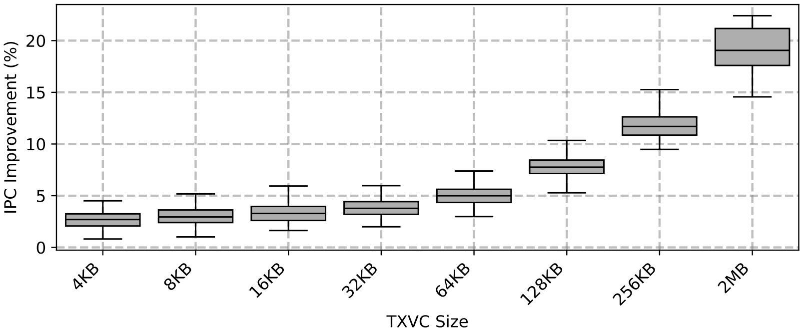
BFF TXVC Size Impact - MPKI



BFF vs SimpleFF

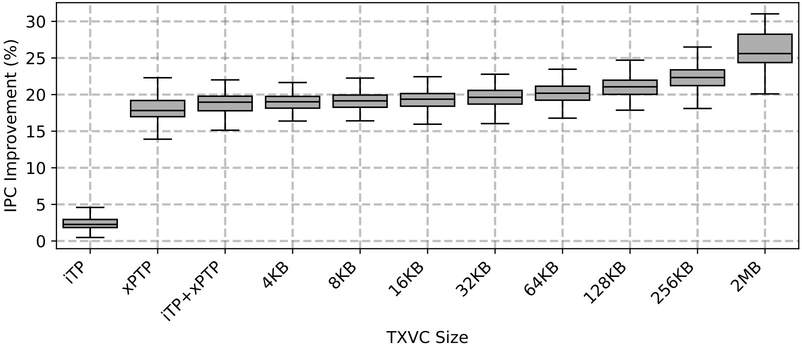


iTP+BFF



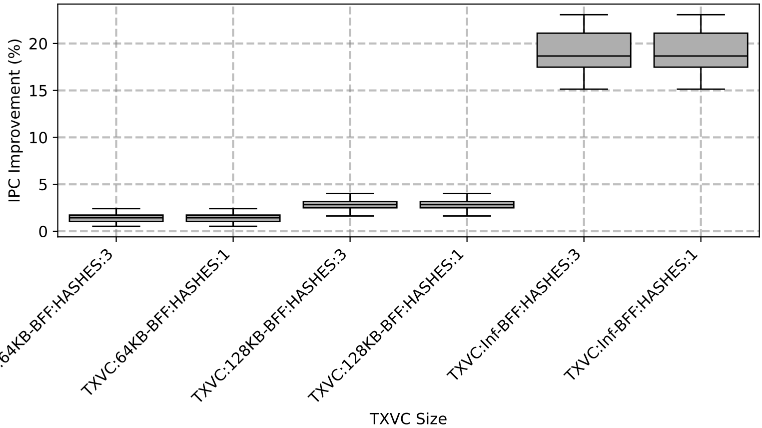
IPC Improvement (%)
20
15
10
5
0
4KB
8KB
16KB
32KB
64KB
128KB
256KB
2MB
TXVC Size

iTP+xPTP+BFF

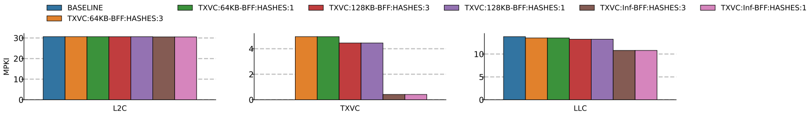


IPC Improvement (%)
0
5
10
15
20
25
30
iTP
xPTP
iTP+xPTP
4KB
8KB
16KB
32KB
64KB
128KB
256KB
2MB
TXVC Size

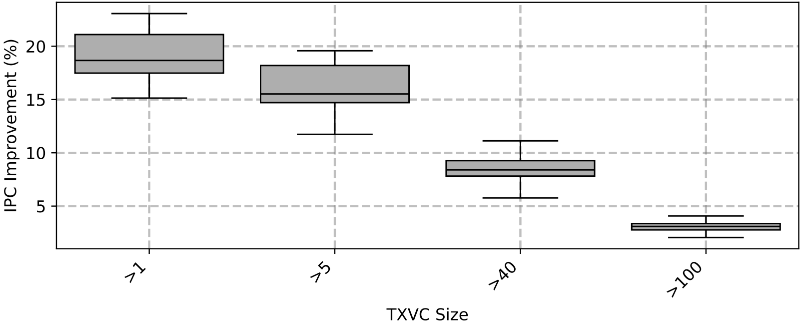
BFF comparison with 1 hash function



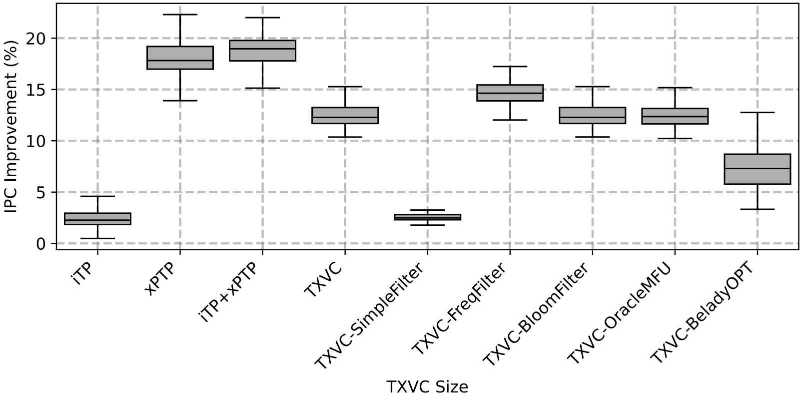
BFF comparison with 1 hash function



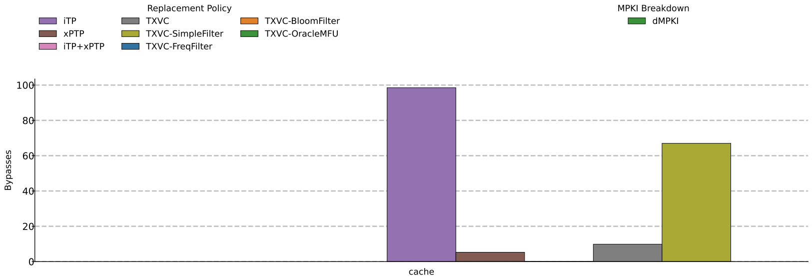
Freq Filter (using a map, no hashes, no size limits)



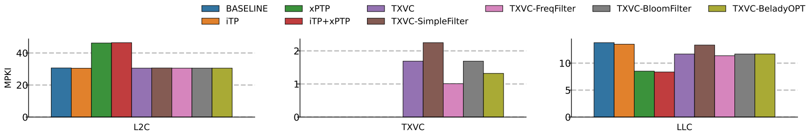
TXVC+Filtering Analysis - 256KB, Full-Assoc, L2C:0.5MB



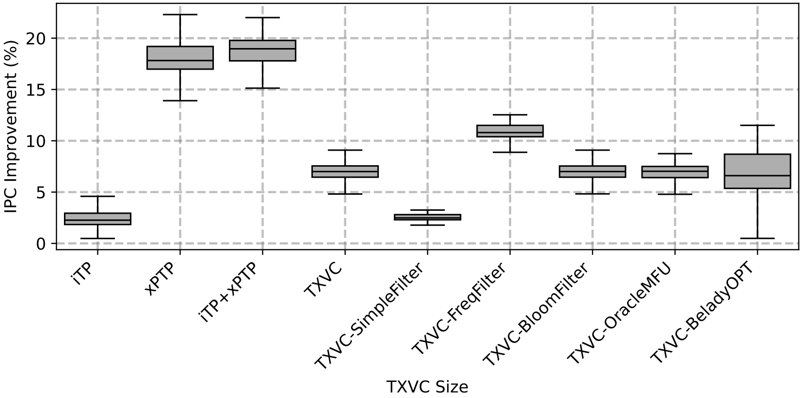
TXVC+Filtering Bypasses - 256KB, Full-Assoc, L2C:0.5MB



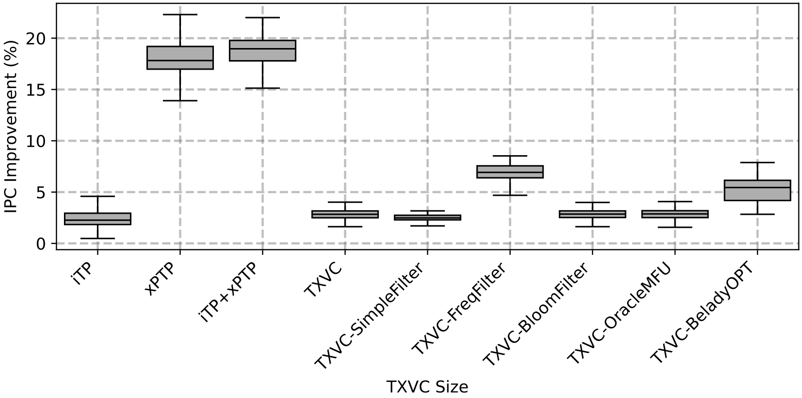
TXVC+Filtering MPKI - 256KB, Full-Assoc, L2C:0.5MB



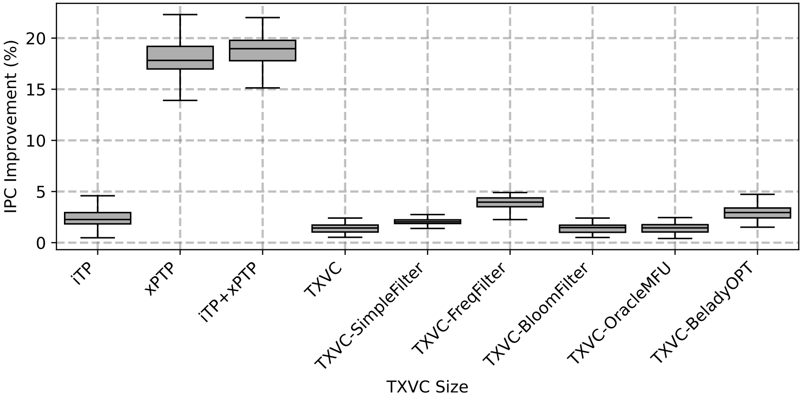
TXVC+Filtering Analysis - 128KB, Full-Assoc, L2C:0.5MB



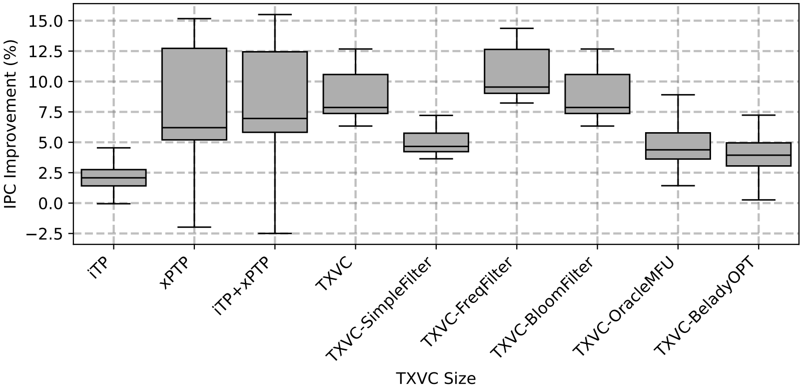
TXVC+Filtering Analysis - 64KB, Full-Assoc, L2C:0.5MB



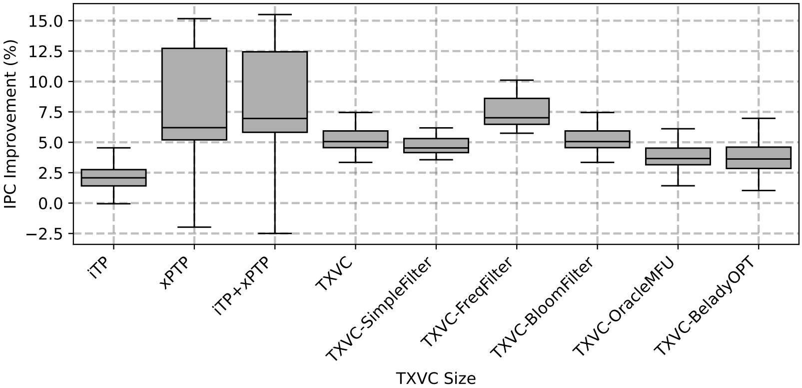
TXVC+Filtering Analysis - 32KB, Full-Assoc, L2C:0.5MB



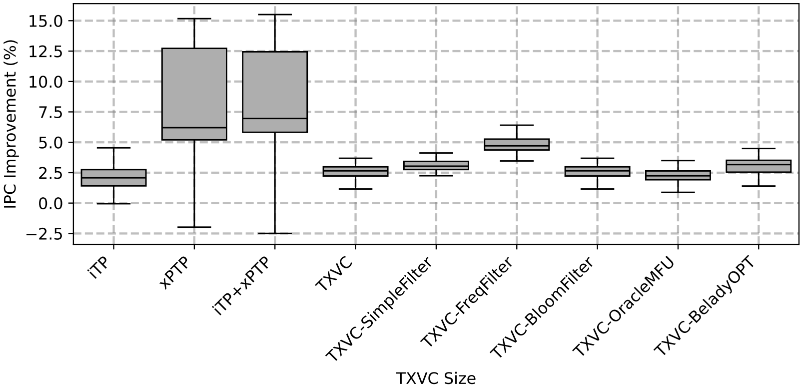
TXVC+Filtering Analysis - 256KB, Full-Assoc, L2C:1MB



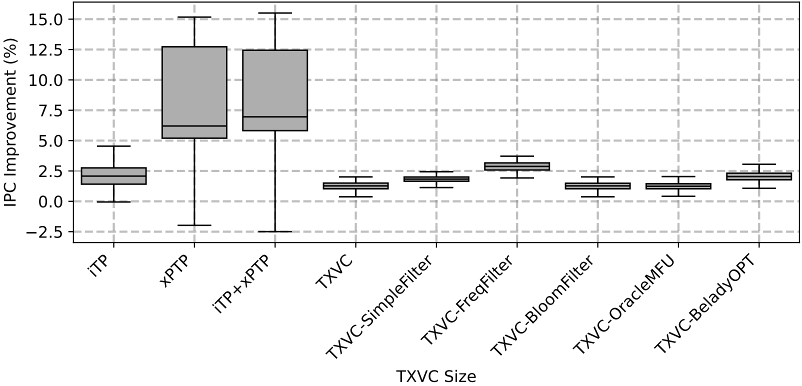
TXVC+Filtering Analysis - 128KB, Full-Assoc, L2C:1MB



TXVC+Filtering Analysis - 64KB, Full-Assoc, L2C:1MB

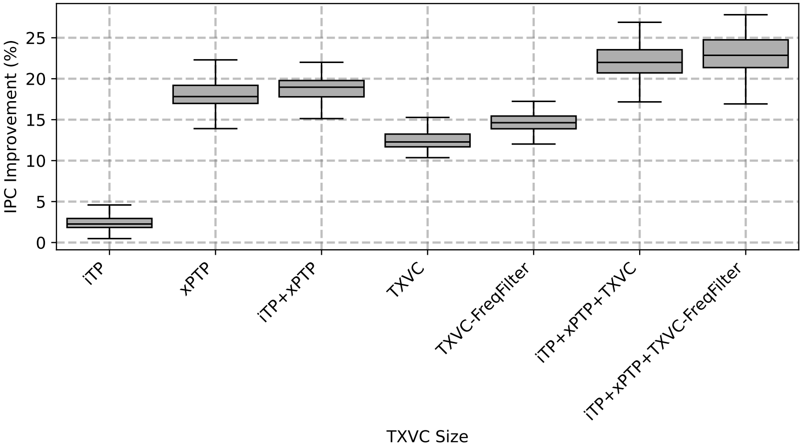


TXVC+Filtering Analysis - 32KB, Full-Assoc, L2C:1MB



iTP+xPTP+TXVC performance

TXVC: 256KB-Full-Assoc, L2C:0.5MB



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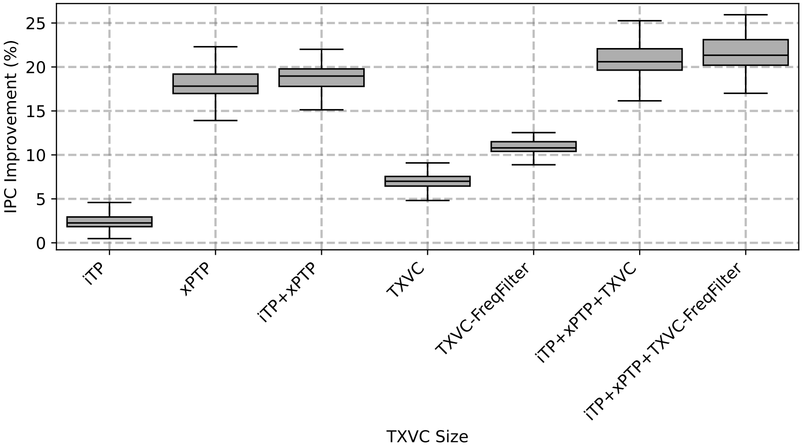

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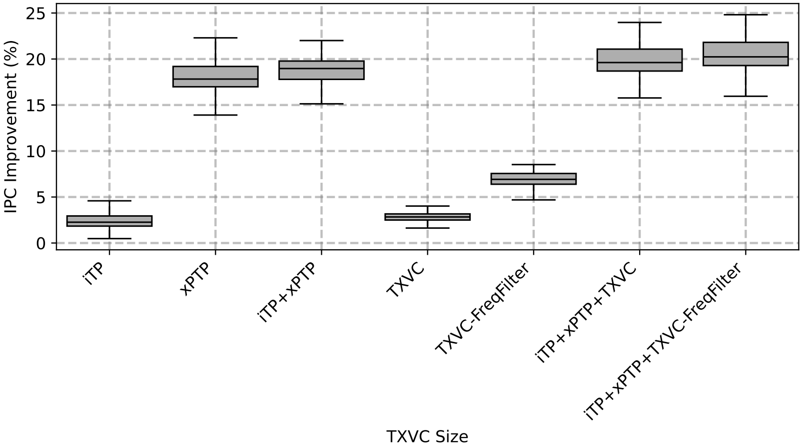

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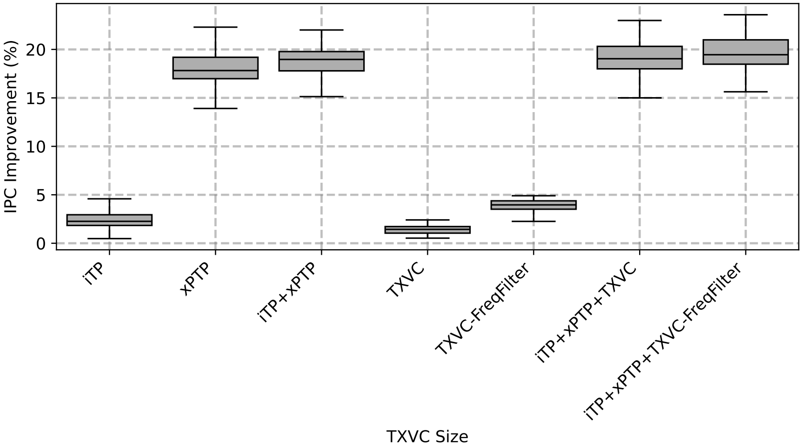

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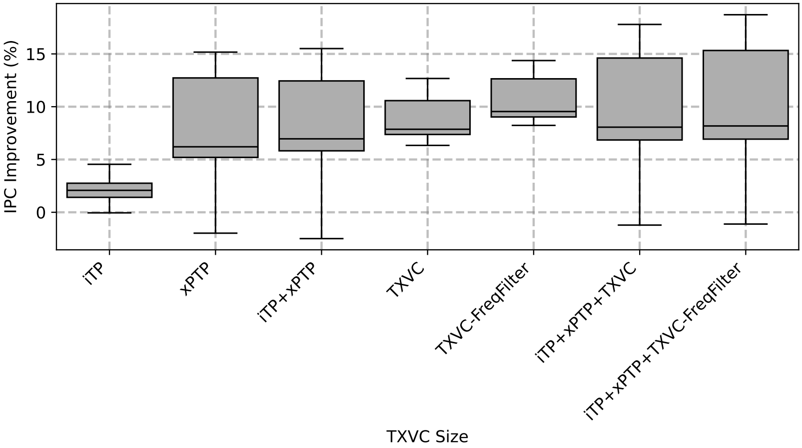

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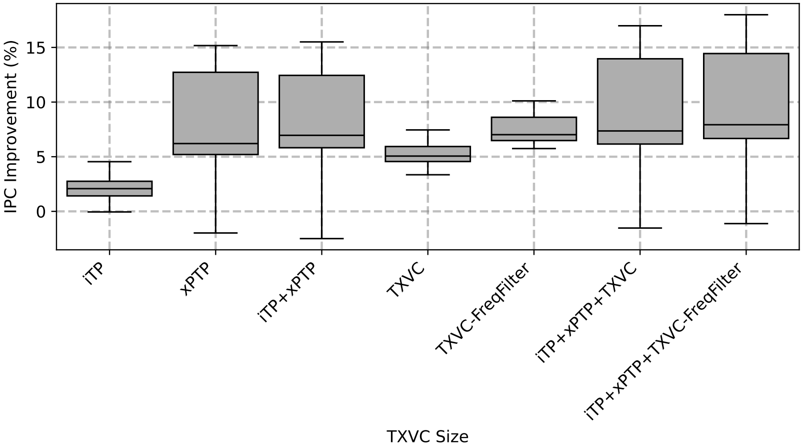

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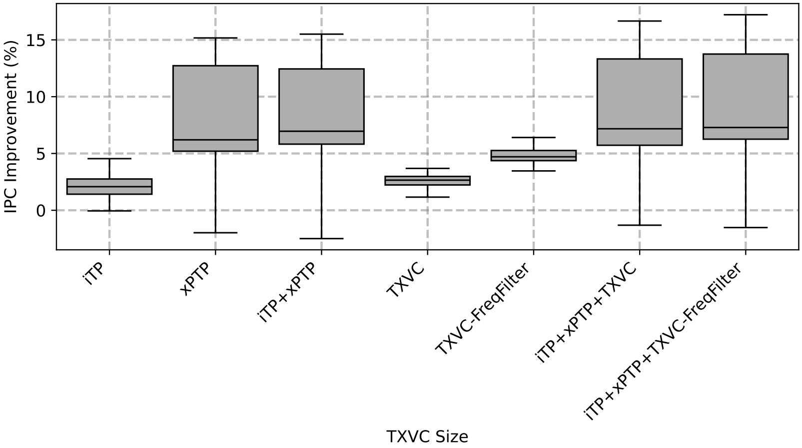

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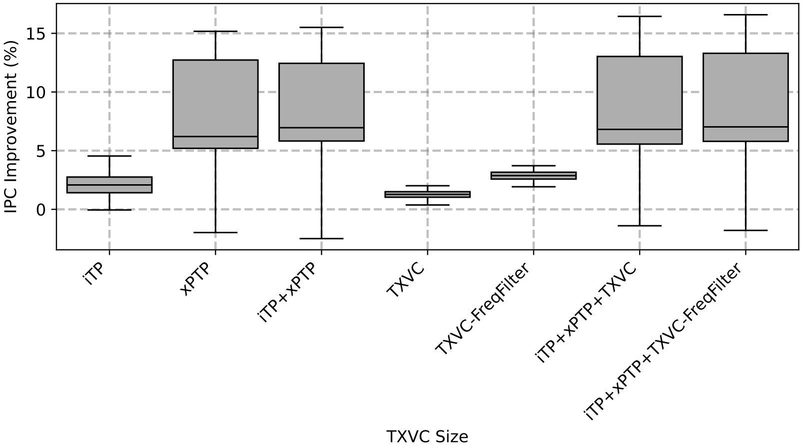
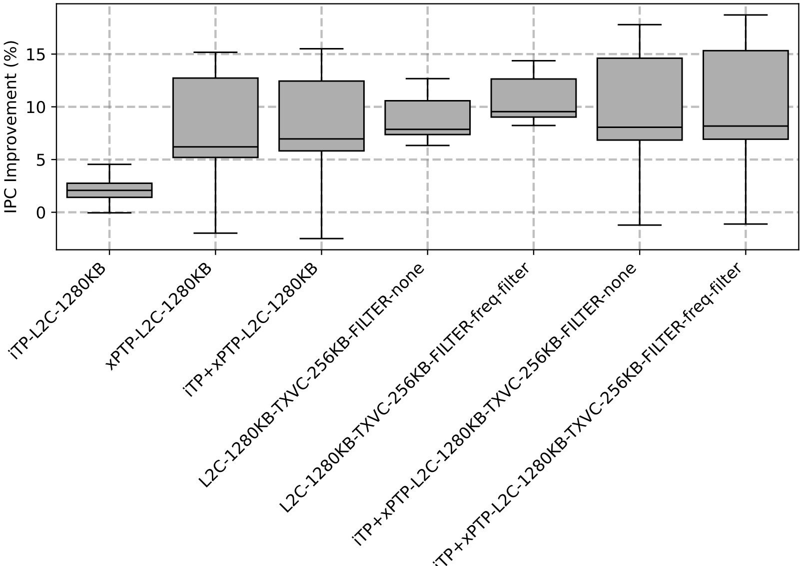
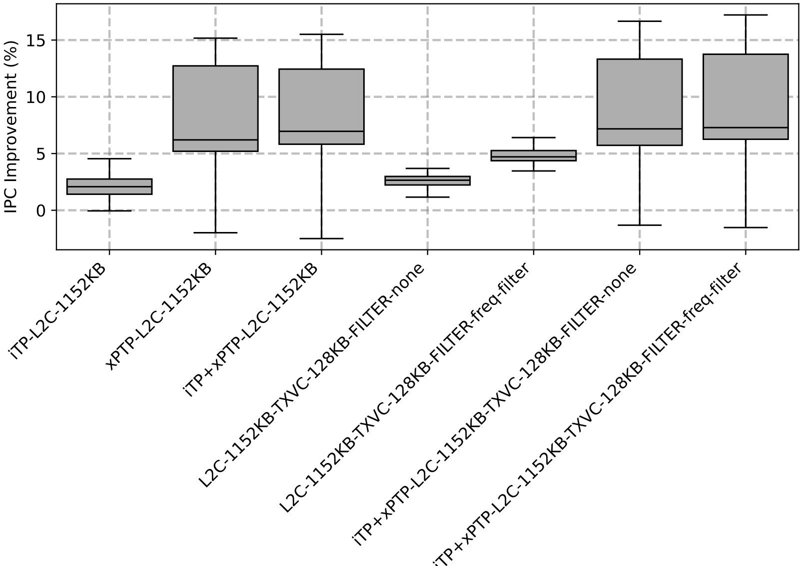


TXVC budget VS L2C budget

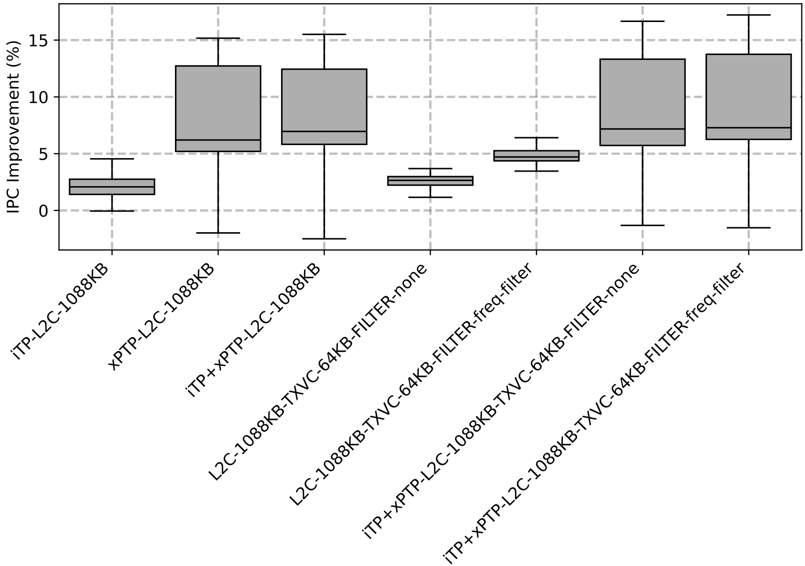
TXVC: 256KB-Full-Assoc, L2C:1280MB-8way



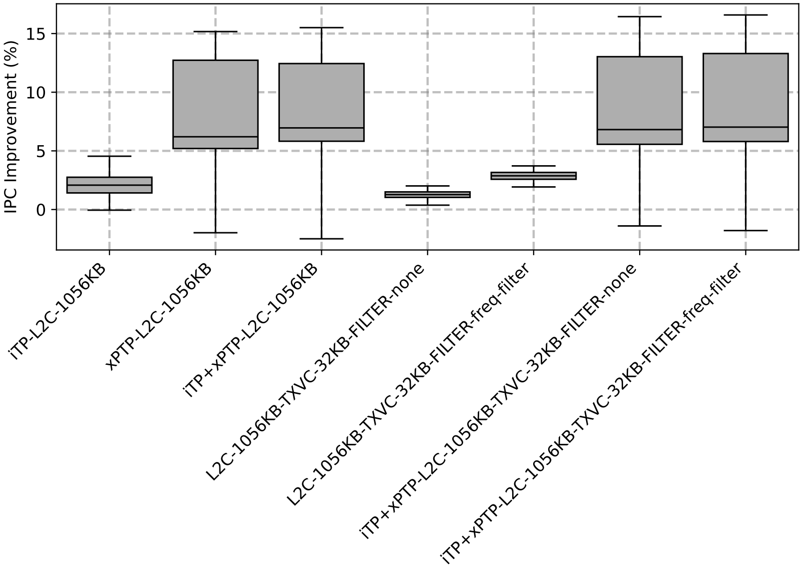
TXVC: 128KB-Full-Assoc, L2C:1152MB-8way



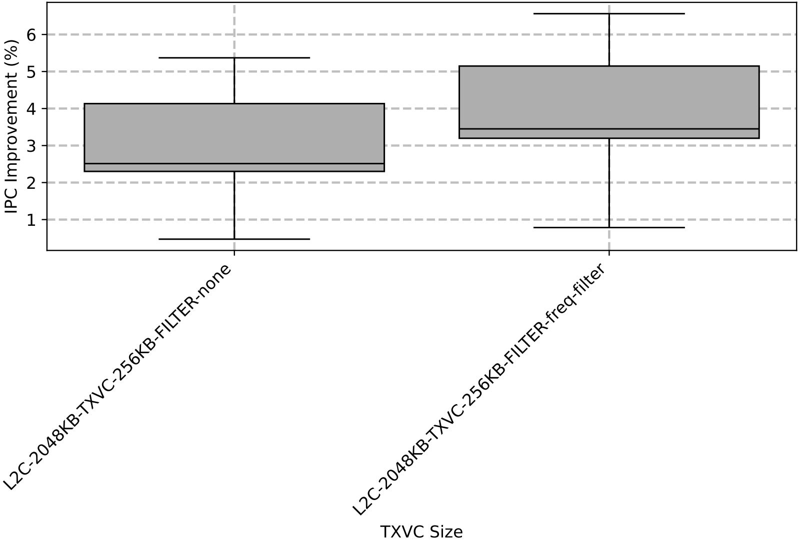
TXVC: 64KB-Full-Assoc, L2C:1088MB-8way



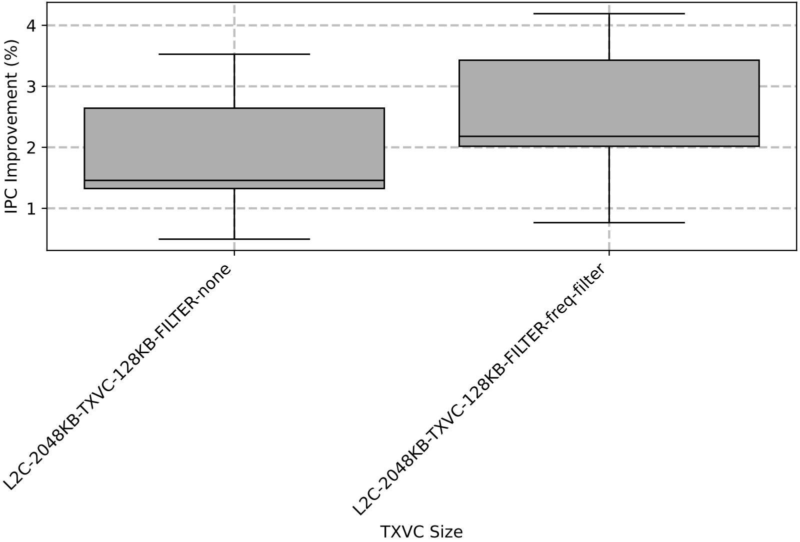
TXVC: 32KB-Full-Assoc, L2C:1056MB-8way



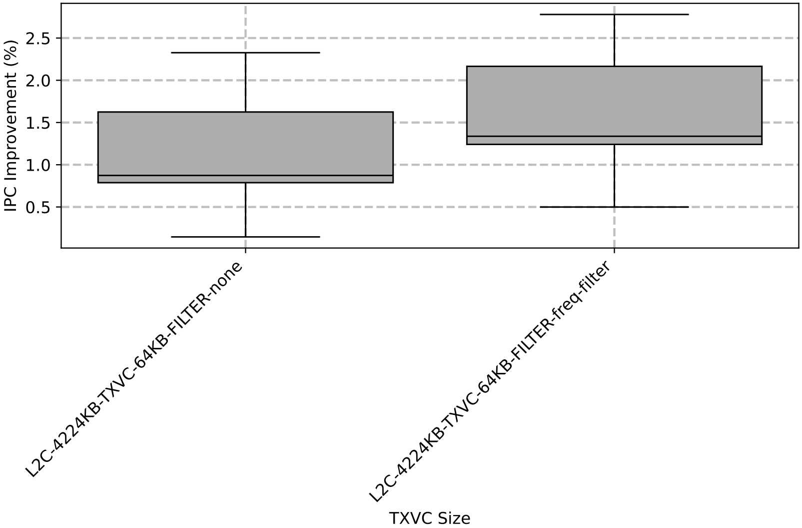
TXVC: 256KB-Full-Assoc, L2C:2304MB-8way



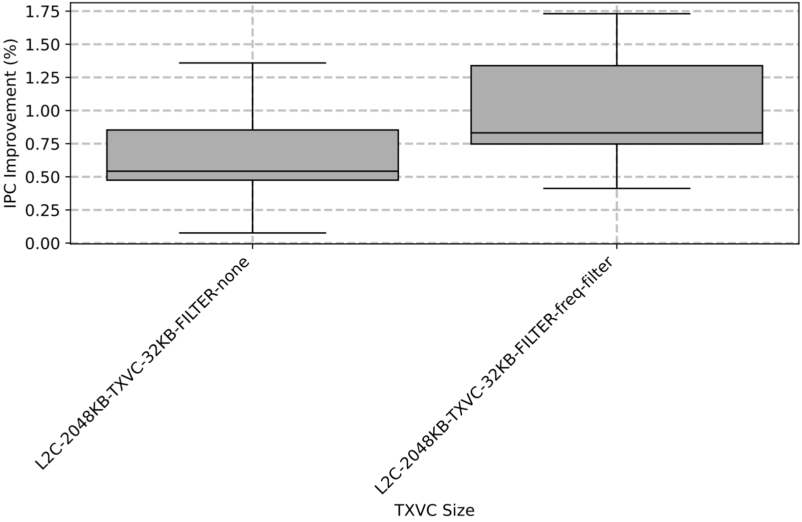
TXVC: 128KB-Full-Assoc, L2C:2176MB-8way



TXVC: 64KB-Full-Assoc, L2C:2112KB-8way



TXVC: 32KB-Full-Assoc, L2C:2080MB-8way



TOP PICKS!!! (iTP)

Top Picks Guideline

- Must demonstrate clear and substantial contributions to computer architecture
  - A three-page (including all references), two-column document using 10-point type
  - First two pages should summarize the paper
  - Third page should argue for the potential of the work to have long-term impact (research, industry) ### lalal
- The final version of the original conference paper
- First document should contain the names of the authors with a footnote that including:
  - The title of the original conference paper
  - The full name of the conference, and date of publication.
- Deadline 20th Dec (notification 1st April)

Accepted Papers Submission

- Micro format
- Must not exceed 6,000 words including no more than:
  - 12 references and short bios of authors
  - Each average-size figure counting as 250 words toward this limit
  - At least 30 percent new content
  - Final papers will be reviewed again before publication (structure, style, clarity, and readability)